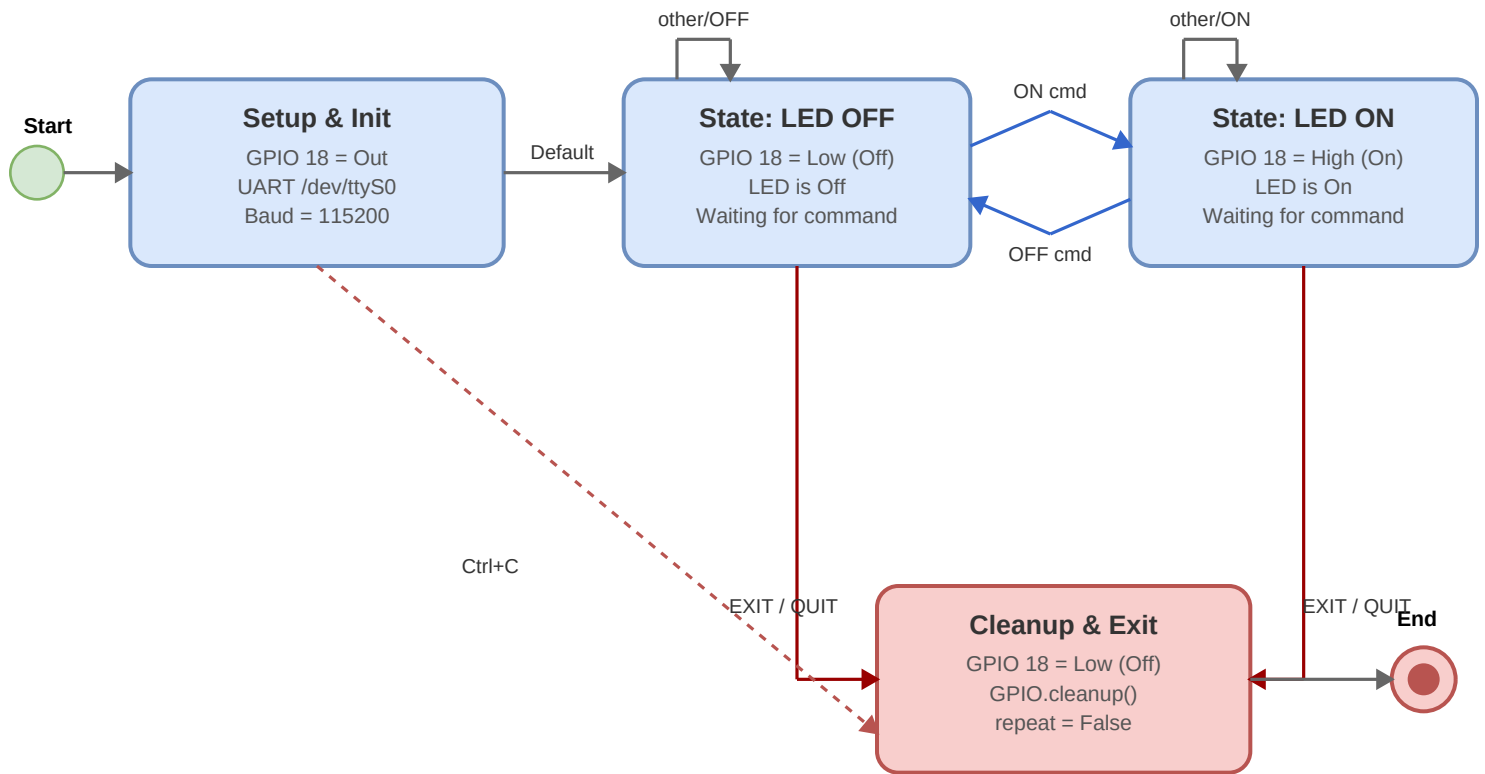


CS 350 Milestone Two: GPIO UART State Machine Diagram

Course: CS-350: Embedded Systems Development | Southern New Hampshire University

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Technical Specifications & Notes:

- The system is modeled as a deterministic Finite State Machine (FSM) running in a client-server paradigm.
- Server operates over physical serial port '/dev/ttyS0' (UART RX) at 115200 baud, 8 data bits, no parity, 1 stop bit.
- GPIO Line 18 is initialized as active output low (Off). Voltage output is 3.3V when enabled (On) to light the physical LED.
- The control loop continuously reads incoming commands via `ser.readline().decode('utf-8').strip().lower()`.
- If command is 'on', the FSM transitions to LED ON and drives GPIO 18 high. If 'off', it transitions to LED OFF and drives GPIO 18 low.
- For 'exit' or 'quit', the FSM transitions to Cleanup & Exit, sets GPIO 18 low, calls `GPIO.cleanup()`, and sets loop variable `repeat = False`.
- A KeyboardInterrupt (Ctrl+C) is handled globally via try-except, routing control instantly to Cleanup & Exit from any active state.